

DSA- SUMMARY

1. Asymptotic Analysis

- a) asymptotic notations
- b) best, avg and worst case
- c) properties of asymptotic notations
- d) graph of big-oh, big-omega, big-theta notations

2. Arrays

- a) Fixed and dynamic size arrays
- b) 1D and 2D arrays
- c) Operations of arrays (searching-linear and binary)
- d) Applications
- e) Advantages and disadvantages
- f) ArrayList in java

3. Linked List (LL)

- a) Comparison with array
- b) Singly linked list (SLL)
- c) Operations like isEmpty(), display() etc
- d) Doubly LL
- e) Advantages over SLL
- f) Circular LL

4. Stack

- a) Algorithm for push, pop operation
- b) Top or peek operation on stack
- c) isEmpty(), isFull() operations
- d) adv and disadv of array-based stack implementation
- e) adv of SLL based stack implementation
- f) expression conversion(6)
- g) recursion
 - a. direct and indirect recursion
 - b. common application
 - c. problem solving using recursion
 - d. tower of Hanoi

5. queue

- a) basic terminologies
- b) queue operations
- c) queue using array
- d) queue using LL
- e) applications
- f) double ended queue (deque)
 - a. operations on deque

- b. deque using DLL
 - c. applications of deque
- g) circular queue
- 6. **priority queues**
 - a) implementation(sorted and unsorted list)
 - b) time complexity
- 7. **map**
 - a) common methods in map
 - b) common implementation
 - c) comparison of 3 implementation
 - d) comparison of dictionary vs map
- 8. **tree**
 - a) linked structure of BT
 - b) array based implementation of BT
 - c) types of BT
 - d) tree traversal algorithms
 - a. inorder
 - b. postorder
 - c. preorder
 - e) binary search tree
 - f) avl trees
 - g) segment trees
 - h) B-tree(insertion)
- 9. **Heap**
 - a) Max and min heap
 - b) Heapify, heapsort
 - c) Heap insertion and deletion
- 10. **Skiplist**
 - a) Operations, implementation
 - b) Time complexity
 - c) Adv and disadv
 - d) Applications
- 11. **External sorting and searching**
- 12. **Graph**
 - a) Types of graphs
 - b) Degree of vertices
 - c) Subgraphs, spanning tree
 - d) Trees and forest
 - e) Graph traversal(DFS and BFS)
 - f) Path and cycle finding
 - g) Graph representation

- h) Dijkstra's algo(single source shortest path)
- i) Minimum spanning tree
 - a. Prim's
 - b. Kruskal's
- j) All pair shortest path
 - a. Floyd warshall algorithm
- k) Topological sorting(digraph)